

Running the Baselined Test Dataset

This is the final run – post all fixes observed via the following approaches:

1. Prompt Engineering
2. Langsmith Observability/Tracing/Evaluators/Datasets and Experiments
3. Fine Tuning Observations

TEST USE CASE	TEST INPUT	TYPE	NOTES	ACTUAL OUTPUT
T1	Generate a project plan for the AI Adoption Project at ABCDE Ltd. Use the provided project description.	Plan (detailed)	Must include phases, milestones, timeline. Must be grounded in the actual project description. Must NOT invent goals not in the description.	 The project plan is for the AI Adoption Project at ABCDE Ltd., which aims to enhance operational efficiency and improve customer experience through the integration of AI-powered tools. This plan is grounded in the project description and task documents retrieved, which outline specific phases, tasks, and key performance indicators such as reducing delivery delays by 15% and lowering operational costs by 10%.  Here's the project plan: ### Phase 1: Project Initiation (31-08-2026 to 08-09-2026) Activities: Define project scope and objectives, Form cross-functional task force Milestones: Project scope documented, Task force established Deliverables: Project scope document, Task force team list ### Phase 2: Data Infrastructure Assessment (31-08-2026 to 22-09-2026) Activities: Audit existing data infrastructure, Identify data gaps and quality issues Milestones: Data audit completed, Data gap analysis report Deliverables: Data infrastructure audit report, Gap analysis report ### Phase 3: Use Case Evaluation (28-09-2026 to 06-10-2026) Activities: Shortlist AI use cases, Vendor evaluation for ML platform, Legal review of data usage Milestones: Use cases shortlisted, Vendor evaluation completed Deliverables: Use case prioritization document, Vendor comparison report, Compliance review document ### Phase 4: Pilot Design (12-10-2026 to 08-11-2026) Activities: Establish data pipelines for AI models Milestones: Data pipelines established Deliverables: Data pipeline architecture document ### Phase 5: Pilot Implementation (09-11-2026 to 06-12-2026) Activities: Implement AI solutions in selected departments, Collect user feedback and performance metrics Milestones: User feedback and performance metrics collected Deliverables: Pilot implementation report, User feedback summary

				### Phase 6: Staff Training (07-12-2026 to 25-01-2027) Activities: Develop and execute staff training program, Complete staff training rollout at remaining sites Milestones: Training program completed, Training extended to all sites Deliverables: Training materials, Training completion report This structured plan aligns with the project's goals and timelines, ensuring that all activities contribute to the overall success metrics of reducing delivery delays by 15% and lowering operational costs by 10%. Automated with this n8n workflow
T2	Generate a project plan for: 'We want to build a chatbot.'	Plan (vague)	Must flag as insufficient detail. Must ask for scope, timeline, team size. Must NOT generate a full plan from one sentence.	🚫 I need a bit more detail to generate a plan. Here's what's missing: specific goals, timeline, and deliverables for the chatbot project. Could you share the project's scope and timeline? Automated with this n8n workflow
T3	Generate a risk assessment for the AI Adoption Project at ABCDE Ltd.	Risk (detailed)	Must produce categorized risks relevant to logistics/AI. Must NOT include generic risks unrelated to project.	⚠️ Here are the relevant risks for the AI Adoption Project at ABCDE Ltd (all already tracked in the register). - Employees resist adopting AI-driven workflows [Change Resistance] Impact: Low adoption and reduced ROI; Mitigation: Strong change management and early engagement - Lack of internal AI/ML expertise [Talent & Skills Gap] Impact: Overreliance on external vendors and slow progress; Mitigation: Provide staff training and hire specialized talent - Difficulty integrating AI with legacy systems [Integration Issues] Impact: Delays in deployment and operational disruptions; Mitigation: Plan integration roadmap and use APIs/middleware - AI models lose accuracy over time [Model Performance Drift] Impact: Incorrect decisions and trust erosion; Mitigation: Regular retraining cycles and monitoring dashboards - Underestimation of AI development costs [Cost Overruns] Impact: Budget constraints and project delays; Mitigation: Set realistic budget and build in contingency funds - Risks of biased models or data privacy violations [Ethical & Legal Concerns] Impact: Legal repercussions and reputational damage; Mitigation: Involve legal early and implement fairness checks - AI models targeted by adversarial attacks [Security Vulnerabilities] Impact: Data breaches and operational risks; Mitigation: Adopt AI security best practices and penetration testing - Incomplete or unstructured data [Data Quality & Availability] Impact: Poor model performance and inaccurate outcomes; Mitigation: Perform early data audits and implement robust data governance Automated with this n8n workflow
T4	Generate a risk assessment for: 'New project starting soon.'	Risk (vague)	Must flag as insufficient. Must ask for project details. Must NOT invent	⚠️ I need more detail to assess risk properly. Here's what's missing: Specific project name or details about the new project..

			project-specific risks from nothing.	Could you share more specifics for the details you are looking for? <i>Automated with this n8n workflow</i>
T5	Generate a weekly status report using the sample task board for Sprint 3.	Status (data)	Must show tasks by status. Must reference actual task names from CSV. Must NOT invent tasks.	 Scoped to Sprint 3 — 4 tasks found, 1 milestone found.  Status report for Sprint 3 (overall health: Green): To do: T009 - Vendor evaluation for ML platform, T025 - Legal review of data usage, T008 - Define success criteria for each use case, T007 - Shortlist AI use cases Upcoming milestones: M05 - Final prioritized use case list with success criteria <i>Automated with this n8n workflow</i>
T6	Generate a weekly status report for: 'Things are going fine.'	Status (filler, no genuine ask) — Assertions	Must ask the user to clarify what they'd like a status update on. Must NOT generate a full report (done/ in_progress/ to_do/ blocked/ at_risk/ upcoming_milestones must all be empty). Must NOT fabricate task statuses or progress percentages.	 Could you share more details on what you'd like a status update on? <i>Automated with this n8n workflow</i>
T7	What are the top 3 risks for the ABCDE Ltd project?	Risk analysis	Must reference actual risks from risk CSV. Must NOT invent risks not in the data.	 Here are the top 3 risks for the ABCDE Ltd project, ranked by severity. - Incomplete or unstructured data [Data Quality & Availability] Impact: Poor model performance and inaccurate outcomes; Mitigation: Perform early data audits and implement robust data governance - Difficulty integrating AI with legacy systems [Integration Issues] Impact: Delays in deployment and operational disruptions; Mitigation: Plan integration roadmap and use APIs/middleware - Lack of internal AI/ML expertise [Talent & Skills Gap] Impact: Overreliance on external vendors and slow progress; Mitigation: Provide staff training and hire specialized talent <i>Automated with this n8n workflow</i>
T8	Which tasks are blocked or at risk of missing their deadline?	Tracking	Must identify T024 (Security review - BLOCKED). Must check due dates against current progress.	 Scoped to All Sprints — 1 task found.  Blocked & At risk — All Sprints: Blocked: T024 - Security review of AI infrastructure (waiting for security team availability) <i>Automated with this n8n workflow</i>
T9	Generate a project plan for a 2-week project.	Plan (edge)	Must ask for scope, goals, deliverables. Must NOT generate detailed plan from timeline alone.	 I need a bit more detail to generate a plan. Here's what's missing: specific project scope, goals, or deliverables. Could you share the project's scope and timeline? <i>Automated with this n8n workflow</i>

T10	Summarize current project status: tasks done, in progress, to do, blocked?	Status summary (Observed Hallucination in Langsmith)	Must count correctly from task board CSV. Counts must match actual data.	 Scoped to All Sprints — 4 done, 23 to do, 1 blocked, 1 at risk, 12 milestones found.  Status report for All Sprints (overall health: Red): Done: T001 - Define project scope and objectives, T002 - Form cross-functional task force, T003 - Initial stakeholder alignment meeting, T004 - Audit existing data infrastructure To do: T006 - Identify data gaps and quality issues, T007 - Shortlist AI use cases, T008 - Define success criteria for each use case, T009 - Vendor evaluation for ML platform, T010 - Collect historical demand data, T011 - Data cleaning and preprocessing pipeline, T012 - Design demand forecasting model, T013 - Design route optimization model, T014 - Build model training pipeline, T015 - Deploy demand model to test environment, T016 - Run pilot with logistics team, T017 - Collect pilot feedback and metrics, T018 - Iterate on model based on feedback, T019 - Prepare production deployment plan, T020 - Staff training program, T021 - Full production rollout, T022 - Set up continuous monitoring, T025 - Legal review of data usage, T026 - Complete staff training rollout (remaining sites), T027 - Full production rollout sign-off (all distribution centers), T028 - Monitor model performance post-launch, T029 - Conduct final project review workshop, T030 - Publish recommendations for Phase 2 scaling Blocked: T024 - Security review of AI infrastructure (BLOCKED: waiting for security team availability) At risk: T005 - Map current logistics workflows (due date 2026-09-19 has passed, not done) Upcoming milestones: M03 - Current systems audit report, M04 - Gap analysis report, M05 - Final prioritized use case list with success criteria, M06 - Data pipelines established, M07 - Working prototype models, M08 - Pilot results report, M09 - User feedback and performance metrics, M10 - Updated models and process refinements, M11 - Full production rollout, M12 - Training sessions completed, M13 - Project closure report, M14 - Recommendations for future scaling <i>Automated with this n8n workflow</i>
T11	What milestones are coming up in the next 2 weeks?	Timeline	Must reference actual milestones from project_timeline.csv. Must NOT invent milestones.	 Upcoming milestones in the next 2 weeks — 3 found.  Status report for All Sprints (overall health: Green): Upcoming milestones: M14 - Recommendations for future scaling, M13 - Project closure report, M29 - Conduct final project review workshop <i>Automated with this n8n workflow</i>
T12	Generate a stakeholder update email for Sprint 2 progress.	Comms	Must reference Sprint 2 tasks. Must be professional. Must NOT include Sprint 3+ tasks.	 Sprint 2 Progress Update Hello, In Sprint 2, the audit of existing data infrastructure (task T004) has been completed successfully. The mapping of current logistics workflows (task T005) is currently in progress, while the identification of data gaps and quality issues (task T006) remains to be done. However, task T005 - Map current logistics workflows (due 2026-09-19, not yet done) has missed its deadline. Next, the team will focus on completing the analysis tasks and preparing for subsequent phases. Best regards, Mira <i>Automated with this n8n workflow</i>
T13	Priya Nair is leaving the organization. Which sprints does this affect?	Risk (detailed)	New Test Case – to accommodate Human in the Loop through Slack approval.	 Priya Nair's departure affects Sprint 2 and Sprint 4 due to her involvement in key tasks and milestones.  Added 2 new risk(s) to the register: R01 - Loss of continuity and expertise in data management and analysis due to Priya Nair's departure. (mitigation: Reassign tasks to other team members and provide necessary training to ensure smooth transition.) R02 - Disruption in the execution of tasks related to data cleaning and preprocessing in Sprint 4. (mitigation: Identify and onboard a replacement for Priya Nair as soon as possible to maintain project momentum.) <i>Automated with this n8n workflow</i>
T14	Arjun Mehta might be stepping back from the project	Risk (detailed)	New Test Case – to accommodate Human in the Loop through Slack approval.	 Arjun Mehta's potential departure could significantly impact several upcoming sprints, particularly Sprint 6, Sprint 9, and Sprint 12, where he is assigned critical tasks.

	soon, what's the impact?			 Added 1 new risk(s) to the register: R12 - Loss of continuity in project tasks due to Arjun Mehta's departure. (mitigation: Reassign tasks to other team members and ensure knowledge transfer before his departure.) <i>Automated with this n8n workflow</i>
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